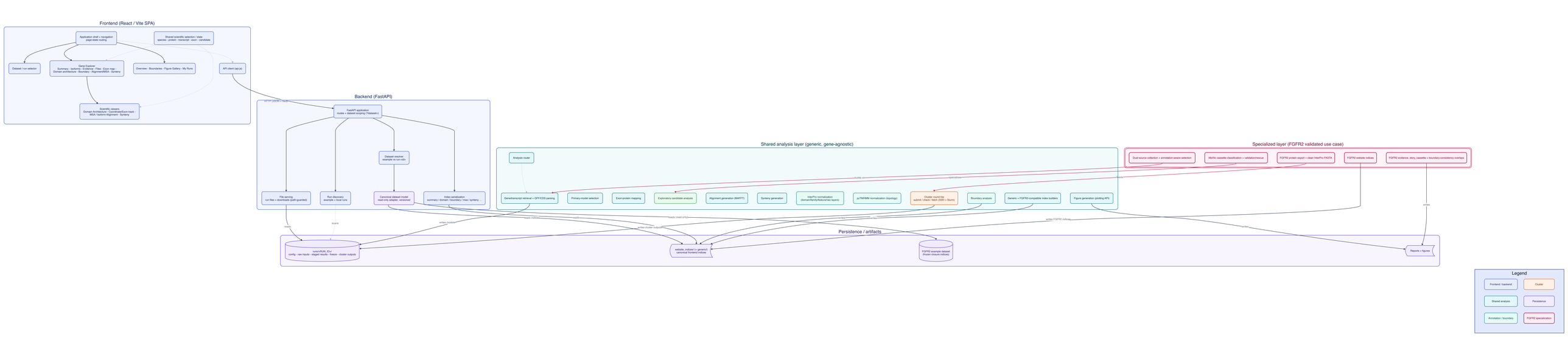

Frontend (React / Vite SPA)
Application shell + navigation
page-state routing
Shared scientific selection / state
species - protein - transcript - exon - candidate
Dataset / run selector
Gene Explorer
Summary - Isoforms - Evidence - Files - Exon map -
Domain architecture - Boundary - Alignment/MSA - Synteny
Overview - Boundaries - Figure Gallery - My Runs
API client (api.js)
Scientific viewers
Domain Architecture - Coordinate/Exon track -
MSA / Isoform Alignment - Synteny
Backend (FastAPI)
HTTP (JSON + files)
FastAPI application
routes + dataset scoping (?dataset=)
Dataset resolver
example vs run:<id>
File serving
run files + downloads (path-guarded)
Run discovery
example + local runs
Canonical dataset model
read-only adapter, versioned
Index serialization
summary / domain / boundary / msa / synteny ...
Shared analysis layer (generic, gene-agnostic)
Analysis router
Gene/transcript retrieval + GFF/CDS parsing
Primary-model selection
Exon-protein mapping
Exploratory candidate analysis
Alignment generation (MAFFT)
Synteny generation
InterPro normalization
(domain/family/feature/raw layers)
pyTMHMM normalization (topology)
Cluster round trip
submit / check / fetch (SSH + Slurm)
Boundary analysis
Generic + FGFR2-compatible index builders
Figure generation (plotting API)
Specialized layer (FGFR2 validated use case)
Dual-source collection + annotation-aware selection
IIIb/IIIc cassette classification + validation/rescue
FGFR2 protein export + clean InterPro FASTA
FGFR2 website indices
FGFR2 evidence, story, cassette + boundary-consistency overlays
builds on
specializes
feeds
reads
scans
writes models
reads indices
reads
reads (read-only)
writes cluster outputs
writes
writes
writes FGFR2 indices
writes
writes
Persistence / artifacts
runs/<RUN_ID>/
config - raw inputs - staged results - freeze - cluster outputs
website_indices/ (+ generic/)
canonical frontend indices
FGFR2 example dataset
(frozen closure indices)
Reports + figures
Legend
Frontend / backend
Cluster
Shared analysis
Persistence
Annotation / boundary
FGFR2 specialization